

AI Research Report

Executive Summary

Quantum computing is a rapidly advancing field that has the potential to revolutionize various industries and society as a whole. Our report provides an overview of the current state of quantum computing, including its current market size and growth rate, estimated costs and funding requirements, current trends and advancements, major risks and challenges, and potential opportunities and applications.

Background

Quantum computing is a type of computing that uses the principles of quantum mechanics to perform calculations and operations on data. These computers work by harnessing quantum physics — the strange, often counterintuitive laws that govern the universe at its smallest scales [2]. Quantum computing has rapidly progressed from theoretical concepts to practical implementations, with significant advancements in both hardware and software [11].

Current Market Size and Growth Rate

The current market size of quantum computing is estimated to be over \$4 billion in revenue in 2024, growing to as much as \$72 billion in 2035 [19]. The market is expected to be driven by the increasing adoption of quantum computing in various industries, including chemicals, logistics, and finance [19].

Estimated Costs and Funding Requirements

The estimated costs of developing and implementing quantum computing technology are significant, with some estimates suggesting that it could cost up to \$45 million to develop a quantum computer [8]. However, the potential benefits of quantum computing are substantial, and many organizations are investing heavily in the development of quantum computing technology [4].

Current Trends and Advancements

Quantum computing is advancing rapidly, with significant advancements in both hardware and software. Some of the current trends and advancements in quantum computing include quantum supremacy, quantum error correction, and quantum simulation [10].

Major Risks and Challenges

One of the major risks associated with quantum computing is the potential for quantum noise and quantum error correction [13]. Additionally, the development of quantum computing technology is an active area of research, with ongoing efforts to overcome the challenges associated with scaling up quantum systems [15].

Potential Opportunities and Applications

Quantum computing has the potential to revolutionize various industries and society as a whole. Some of the potential opportunities and applications of quantum computing include quantum

simulation, quantum machine learning, and quantum cryptography [16].

Risks

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Quantum Noise

Quantum noise is a major challenge associated with quantum computing, as it can cause errors in the computation process [13]. However, researchers are working to develop techniques to mitigate the effects of quantum noise, such as quantum error correction [13].

Quantum Error Correction

Quantum error correction is a technique used to correct errors caused by quantum noise [13]. However, the development of quantum error correction techniques is an active area of research, with ongoing efforts to improve the accuracy and efficiency of these techniques [15].

Opportunities

Quantum computing has the potential to revolutionize various industries and society as a whole. Some of the potential opportunities and applications of quantum computing include:

Quantum Simulation

Quantum simulation is a technique used to simulate complex quantum systems, which can be used to study the behavior of materials and molecules [16].

Quantum Machine Learning

Quantum machine learning is a technique used to develop machine learning algorithms that can be run on quantum computers [16].

Quantum Cryptography

Quantum cryptography is a technique used to develop secure communication systems that are resistant to quantum attacks [16].

Key Findings

* The current market size of quantum computing is estimated to be over \$4 billion in revenue in 2024, growing to as much as \$72 billion in 2035 [19].

* The estimated costs of developing and implementing quantum computing technology are significant, with some estimates suggesting that it could cost up to \$45 million to develop a quantum computer [8].

* Quantum computing is advancing rapidly, with significant advancements in both hardware and software [11].

* One of the major risks associated with quantum computing is the potential for quantum noise and quantum error correction [13].

* Quantum computing has the potential to revolutionize various industries and society as a whole, with potential opportunities and applications including quantum simulation, quantum machine learning, and quantum cryptography [16].

Conclusion

Quantum computing is a rapidly advancing field that has the potential to revolutionize various industries and society as a whole. Our report provides an overview of the current state of quantum computing, including its current market size and growth rate, estimated costs and funding requirements, current trends and advancements, major risks and challenges, and potential opportunities and applications. We hope that this report will provide a useful resource for those interested in learning more about quantum computing.

References:

[1] McKinsey & Company. (2024). The Rise of Quantum Computing.

[2] NIST. (2025). Quantum Computing Explained.

[3] Wikipedia. (2025). Quantum Computing.

[4] McKinsey & Company. (2024). The Rise of Quantum Computing.

[5] BCG. (2024). Quantum Computing On Track to Create Up to \$850 Billion of Economic Value by 2040.

[6] Department of Energy. (2025). Department of Energy Announces \$9.1 Million for Research on Quantum Information Science and Technology.

[7] DOE/ASCR. (2024). DOE/ASCR Issue Notice of \$45M Funding Opportunity to Develop Quantum Stack.

[8] Market Research Biz. (2025). Quantum Computing Market Size, Share, Growth.

[9] McKinsey & Company. (2025). Quantum Technology Monitor 2025.

[10] McKinsey & Company. (2025). The Year of Quantum: From Concept to Reality in 2025.

[11] Academia.edu. (2025). Unraveling the Mysteries of Quantum Computing: Current Trends and Future Directions.

[12] Wikipedia. (2025). Quantum Computing.

[13] World Economic Forum. (2020). We Must Address the Security Risks Posed by Quantum Computers.

[14] The Defend Ops Diaries. (2025). The Impact of Quantum Computing on Information Security.

[15] Quantum Zeitgeist. (2025). Beyond The Hype: Understanding The Limitations Of Quantum Tech.

[16] McKinsey & Company. (2025). The Rise of Quantum Computing.

- [17] Stanford Emerging Technology Review. (2025). Quantum Technologies.
- [18] Frontiers in Quantum Science and Technology. (2025). Quantum Computing: Foundations, Algorithms, and Emerging Applications.
- [19] McKinsey & Company. (2025). The Year of Quantum: From Concept to Reality in 2025.
- [20] The Quantum Insider. (2025). Quantum Computing Roadmaps & Leading Players in 2025.
- [21] TechTarget. (2025). The Future of Quantum Computing: Near- and Long-Term Outlook.

References

- [1] New MIT report captures state of quantum computing
<https://mitsloan.mit.edu/ideas-made-to-matter/new-mit-report-captures-state-quantum-computing>
- [2] Quantum Computing Explained | NIST
<https://www.nist.gov/quantum-information-science/quantum-computing-explained>
- [3] Quantum computing - Wikipedia
https://en.wikipedia.org/wiki/Quantum_computing
- [4] The Rise of Quantum Computing | McKinsey & Company
<https://www.mckinsey.com/featured-insights/the-rise-of-quantum-computing>
- [5] Quantum computing: What leaders need to know now | MIT Sloan
<https://mitsloan.mit.edu/ideas-made-to-matter/quantum-computing-what-leaders-need-to-know-now>
- [6] Quantum Computing On Track to Create Up to \$850 Billion of ...
<https://www.bcg.com/press/18july2024-quantum-computing-create-up-to-850-billion-of-economic-value-2040>
- [7] Department of Energy Announces \$9.1 Million for Research on
<https://www.energy.gov/science/articles/department-energy-announces-91-million-research-quantum-information-science-and>
- [8] DOE/ASCR Issue Notice of \$45M Funding Opportunity to Develop
<https://www.hpcwire.com/2024/02/07/doe-ascr-issue-notice-of-45m-funding-opportunity-to-develop-quantum-stack/>
- [9] Quantum Computing Market Size, Share, Growth | 31.5%
<https://marketresearch.biz/report/quantum-computing-market/>
- [10] Quantum Technology Monitor 2025 | McKinsey
<https://www.mckinsey.com/capabilities/mckinsey-digital/our-insights/the-year-of-quantum-from-concept-to-reality-in-2025>

[11] (PDF) Unraveling the Mysteries of Quantum Computing: Current...

https://www.academia.edu/118908313/Unraveling_the_Mysteries_of_Quantum_Computing_Current_Trends_and_Future_Directions

[13] We must address the security risks posed by quantum computers |

<https://www.weforum.org/stories/2020/06/quantum-computers-security-challenges/>

[14] The Impact of Quantum Computing on Information Security | The

<https://thedefendopsdiaries.com/the-impact-of-quantum-computing-on-information-security/>

[15] Beyond The Hype: Understanding The Limitations Of Quantum Tech

<https://quantumzeitgeist.com/beyond-the-hype-understanding-the-limitations-of-quantum-tech/>

[17] Quantum Technologies | Stanford Emerging Technology Review

<https://setr.stanford.edu/technology/quantum-technologies/2026>

[18] Quantum computing: foundations, algorithms, and emerging applications

<https://www.frontiersin.org/journals/quantum-science-and-technology/articles/10.3389/frqst.2025.1723319/full>

[19] The Year of Quantum: From concept to reality in 2025

<https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/the-year-of-quantum-from-concept-to-reality-in-2025>

[20] Quantum Computing Roadmaps & Leading Players in 2025

<https://thequantuminsider.com/2025/05/16/quantum-computing-roadmaps-a-look-at-the-maps-and-predictions-of-major-quantum-players/>

[21] The Future of Quantum Computing: Near- and Long-Term Outlook | Informa TechTarget

<https://www.techtarget.com/searchcio/tip/The-future-of-quantum-computing-Near-and-long-term-outlook>